

POLARIS

Version 1.0

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1. Introduction

1.1 Overview

POLARIS (Platform for Online Literature Analysis, Retrieval, and Integrated Search) is an R/Shiny-based application designed to support evidence synthesis and systematic literature reviews through integrated searches across multiple scientific databases. The platform was developed to streamline the literature search process by providing a unified interface capable of querying multiple databases simultaneously, monitoring search progress in real time, organizing retrieved records, and exporting results in formats compatible with reference management software and review platforms.

POLARIS reduces the need to perform independent searches in multiple databases and promotes transparency, reproducibility, and efficiency in literature review workflows.

1.2 Objectives

The primary objectives of POLARIS are:

- Facilitate comprehensive literature searches.
- Integrate multiple bibliographic databases into a single workflow.
- Support systematic reviews and evidence synthesis.
- Improve transparency and reproducibility.
- Reduce manual effort during search and retrieval stages.
- Generate exportable reference files for downstream screening and analysis.

1.3 Intended Users

POLARIS is intended for:

- Researchers
 - Graduate students
 - Faculty members
 - Evidence synthesis teams
 - Systematic review practitioners
 - Public health researchers
 - Environmental and climate scientists
 - Social scientists
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2. System Architecture

POLARIS was developed using the R programming language and the Shiny framework.

The application combines:

- Graphical user interface (GUI)
- Parallelized search execution
- API-based retrieval
- Export management system
- Monitoring dashboard

Searches are executed through dedicated worker processes, allowing users to continue interacting with the application while records are being retrieved.

3. Supported Databases

3.1 PubMed

PubMed is accessed through the National Center for Biotechnology Information (NCBI) API.

Retrieved information may include:

- Title
- Authors
- Journal
- Abstract
- Publication date
- PMID
- DOI

3.2 Scopus

Scopus records are retrieved through Elsevier APIs.

Available metadata may include:

- Title
- Authors
- Journal
- Abstract
- Keywords
- Citation information
- DOI

3.3 ScienceDirect

ScienceDirect retrieval is performed through Elsevier services.

Retrieved records may include:

- Title
- Authors
- Journal
- Abstract
- Keywords
- Citation information
- DOI

3.4 Semantic Scholar

Semantic Scholar searches are performed through the Semantic Scholar Academic Graph API.

Available metadata may include:

- Title
- Authors
- Venue
- Abstract
- Publication year
- Citation count

- DOI
 - Fields of study
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4. Starting POLARIS

4.1 Launching the Application

After installing all required dependencies, launch POLARIS by executing:

```
shiny::runApp()
```

The application will open in the default web browser.

4.2 Home Screen

The main interface contains:

- Search configuration panel
 - Database selection menu
 - Search execution controls
 - Monitoring dashboard
 - Results viewer
 - Export tools
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5. Conducting a Literature Search

5.1 Building a Query

Users may enter:

- Keywords
- Boolean operators
- Phrase searches
- Field-specific expressions

Examples:

("street vendors" OR "street vending")

AND

(heat OR temperature OR thermal stress)

("urban heat island")

AND

(vulnerability OR exposure)

5.2 Selecting Databases

Users may choose one or more databases simultaneously.

Recommended practice:

- Broad searches: all databases
- Medical topics: PubMed + Scopus
- Interdisciplinary topics: all databases
- Rapid searches: Semantic Scholar + PubMed

5.3 Running Searches

After selecting databases and entering a query:

1. Click Search.
2. POLARIS initializes retrieval workers.
3. Queries are submitted to selected databases.
4. Results are retrieved incrementally.

5. Progress is displayed in real time.

6. Monitoring Dashboard

The monitoring dashboard provides live feedback regarding search execution.

6.1 Search Status

Displayed information may include:

- Database currently being queried
- Number of retrieved records
- Execution progress
- Completion status

6.2 Runtime Logs

The logging system records:

- Search initiation
- API requests
- Pagination events
- Export events
- Warnings
- Errors

Logs facilitate troubleshooting and reproducibility.

7. Retrieval Workflow

POLARIS follows the workflow below:

1. User submits query.
2. Query is validated.
3. Database-specific requests are generated.
4. APIs are contacted.
5. Records are retrieved.
6. Pagination procedures are executed when required.
7. Results are consolidated.
8. Data are stored locally.
9. Exports become available.

8. Result Management

8.1 Viewing Results

Retrieved records may be reviewed directly within the application.

Information typically displayed includes:

- Article title
- Authors
- Journal
- Publication year
- DOI
- Source database

8.2 Organizing Results

POLARIS automatically stores search outputs in designated folders.

Folder organization may include:

- Search date
- Database source
- Search identifier
- Export type

This structure supports project management and reproducibility.

9. Exporting Records

9.1 RIS Export

POLARIS exports bibliographic records in RIS format.

RIS files can be imported into:

- Zotero
- EndNote
- Mendeley
- Rayyan
- Covidence

9.2 Export Procedure

1. Complete a search.
 2. Select Export.
 3. Choose RIS format.
 4. Save the generated file.
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10. Reproducibility Features

POLARIS was designed to support transparent evidence synthesis.

The platform records:

- Search date
- Search query
- Selected databases
- Retrieved record counts
- Export history
- Runtime logs

These records facilitate reporting according to systematic review guidelines.

11. Best Practices

To maximize search quality:

- Use controlled vocabulary when available.
 - Apply Boolean operators carefully.
 - Search multiple databases.
 - Save search strategies.
 - Maintain exported RIS files.
 - Document all search decisions.
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12. Limitations

Users should be aware that:

- API availability depends on provider services.
- Search results may change over time as databases are updated.
- Rate limits imposed by providers may affect retrieval speed.
- Metadata completeness varies among databases.

13. Future Developments

Planned enhancements include:

- Search validation module
- Precision and recall assessment
- Screening workflow integration
- Expanded export formats
- Additional bibliographic databases
- Advanced search analytics

14. Citation

If POLARIS contributes to a publication, users should cite the software as:

Prado, L., Vanos, J., & Rosales Chavez, J. B. (2026). Polaris - A multi-database literature review system. Zenodo. <https://doi.org/10.5281/zenodo.19896742>

15. Support

For questions, bug reports, or feature requests, please contact the software developer.

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